

Macroeconomics III
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Chapter 5

TECHNOLOGICAL
PROGRESS AND
GROWTH: THE
GENERAL SOLOW
MODEL

THE GENERAL SOLOW MODEL

- Back to a closed economy
- In the basic Solow model: no growth in GDP per worker in steady state. This contradicts the empirics for the Western world (stylized fact #5). In the general Solow model:
- Total factor productivity, B_t , is assumed to grow at a constant, exogenous rate (the only change). This implies a steady state with balanced growth and a constant, positive growth rate of GDP per worker.
- The source of long-run growth in GDP per worker in this model is **exogenous** technological growth. Not deep, but:
 - It's not trivial that the result is **balanced** growth in steady state.
 - It's reassuring for applications that the model is in accordance with a fundamental empirical regularity.
- Our focus is still:
what creates economic progress and prosperity...

THE MICRO WORLD OF THE SOLOW MODEL

... is the same as in the basic Solow model, e.g.,

- The same **object** (a closed economy)
- The same **goods** and **markets**. Once again, markets are competitive with real prices of 1, r_t and w_t , respectively. There is one type of output (one sector model).
- The same **agents**: consumers and firms (and governments), essentially with the same behaviour, specifically: one representative profit-maximizing firm decides K_t^d and L_t^d given r_t and w_t .
- One difference: **the production function**. There is a possibility of technological progress:

$$Y_t = B_t K_t^\alpha L_t^{1-\alpha}, \quad 0 < \alpha < 1.$$

The full sequence (B_t) is exogenous and $B_t > 0$ for all t .
Special case is $B_t = B$ (basic Solow model).

THE PRODUCTION FUNCTION WITH TECHNOLOGICAL PROGRESS

$Y_t = B_t K_t^\alpha L_t^{1-\alpha}$ with a given sequence, $(B_t) \Leftrightarrow$

$Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}$ with a given sequence, (A_t) , $A_t \equiv B_t^{1/(1-\alpha)}$.

- With a Cobb-Douglas production function it makes no difference whether we describe technical progress by a sequence, (B_t) , for TFP or by the corresponding sequence, (A_t) , for labour augmenting productivity.
- In our case, the latter is the most convenient. The exogenous sequence, (A_t) , is given by:

$$A_{t+1} = (1 + g) A_t, \quad g > -1$$

$$\Rightarrow A_t = (1 + g)^t A_0, \quad g > -1$$

- Technical progress comes as “manna from heaven” (it requires no input of production).

- Remember the definitions: $y_t \equiv Y_t / L_t$ and $k_t \equiv K_t / L_t$.
- Dividing by L_t on both sides of $Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}$ gives the per capita production function:

$$y_t = k_t^\alpha A_t^{1-\alpha}.$$

- From this follows:

$$\ln y_t - \ln y_{t-1} = \alpha (\ln k_t - \ln k_{t-1}) + (1-\alpha) (\ln A_t - \ln A_{t-1}) \Leftrightarrow$$

$$g_t^y = \alpha g_t^k + (1-\alpha) g_t^A \cong \alpha g_t^k + (1-\alpha) g.$$

Growth in y_t can come from exactly two sources, and g_t^y is the weighted average of g_t^k and g with weights α and $1-\alpha$.

- If, as in balanced growth, k_t / y_t is constant, then $g_t^y = g$!

THE COMPLETE SOLOW MODEL

$$Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}$$

$$r_t = \alpha K_t^{\alpha-1} (A_t L_t)^{1-\alpha} = \alpha \left(\frac{K_t}{A_t L_t} \right)^{\alpha-1}$$

$$w_t = (1-\alpha) K_t^\alpha L_t^{-\alpha} A_t^{1-\alpha} = (1-\alpha) \left(\frac{K_t}{A_t L_t} \right)^\alpha A_t$$

$$S_t = s Y_t$$

$$K_{t+1} - K_t = S_t - \delta K_t, \quad K_0 \text{ given}$$

$$L_{t+1} = (1+n) L_t, \quad L_0 \text{ given}$$

$$A_{t+1} = (1+g) A_t, \quad A_0 \text{ given}$$

- Parameters: α, s, δ, n, g . Let $g > 0$.
- State variables: K_t, L_t and A_t .
- Full model? Yes, given K_0, L_0 and A_0 the model determines the full sequences $(K_t), (L_t), (A_t), (Y_t), (r_t), (w_t), (S_t)$.

- **Note:**

$$r_t K_t = \alpha K_t^\alpha (A_t L_t)^{1-\alpha} = \alpha Y_t$$
$$w_t L_t = (1-\alpha) K_t^\alpha (A_t L_t)^{1-\alpha} = (1-\alpha) Y_t.$$

That is: capital's share = α , labour's share = $1-\alpha$, pure profits = 0 . Our α should still be around $1/3$.

- **Also note:** defining "effective labour input" as $\tilde{L}_t = A_t L_t$:

$$\tilde{L}_{t+1} = (1+n)(1+g)\tilde{L}_t \equiv (1+\tilde{n})\tilde{L}_t,$$

The model is mathematically equivalent to the basic Solow model with $\tilde{L}_t$ taking the place of L_t , and $\tilde{n}$ taking the place of n , and with $B=1$!

We could, in principle, take over the full analysis from the basic Solow model, but we will nevertheless be...

ANALYSING THE GENERAL SOLOW MODEL

- **If** the model implies convergence to a steady state with balanced growth, **then** in steady state k_t and y_t must grow at the same constant rate (recall again that k_t / y_t is constant under balanced growth). Remember also:

$$g_t^y = \alpha g_t^k + (1 - \alpha) g_t^A.$$

Hence, if $g_t^y = g_t^k$, then $g_t^y = g_t^k = g_t^A$. **If** there is convergence towards a steady state with balanced growth, **then** in this steady state k_t and y_t will both grow at the same rate as A_t , and hence k_t / A_t and y_t / A_t will be constant.

- **Furthermore**, from the aforementioned equivalence to the basic Solow model, $K_t / \tilde{L}_t = K_t / (A_t L_t) = k_t / A_t$ and $Y_t / \tilde{L}_t = Y_t / (A_t L_t) = y_t / A_t$ converge towards constant steady state values.
- Each of the above observations suggests analysing the model in terms of:

1. $\tilde{k}_t \equiv \frac{k_t}{A_t} = \frac{K_t}{A_t L_t}$ and $\tilde{y}_t \equiv \frac{y_t}{A_t} = \frac{Y_t}{A_t L_t}$.

2. From $Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}$ we get $\tilde{y}_t = \tilde{k}_t^\alpha$.

3. From $K_{t+1} - K_t = S_t - \delta K_t$ and $S_t = sY_t$ we get

$$K_{t+1} = sY_t + (1 - \delta)K_t$$

4. Dividing by $A_{t+1}L_{t+1} = (1+g)(1+n)A_tL_t$ on both sides gives

$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g)} \left(s\tilde{y}_t + (1-\delta)\tilde{k}_t \right).$$

5. Inserting $\tilde{y}_t = \tilde{k}_t^\alpha$ gives the **transition equation**:

$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g)} \left(s\tilde{k}_t^\alpha + (1-\delta)\tilde{k}_t \right).$$

6. Subtracting $\tilde{k}_t$ from both sides gives the **Solow equation**:

$$\tilde{k}_{t+1} - \tilde{k}_t = \frac{1}{(1+n)(1+g)} \left(s\tilde{k}_t^\alpha - (n+g+\delta+ng)\tilde{k}_t \right).$$

CONVERGENCE TO STEADY STATE: THE TRANSITION DIAGRAM

- The transition equation is:

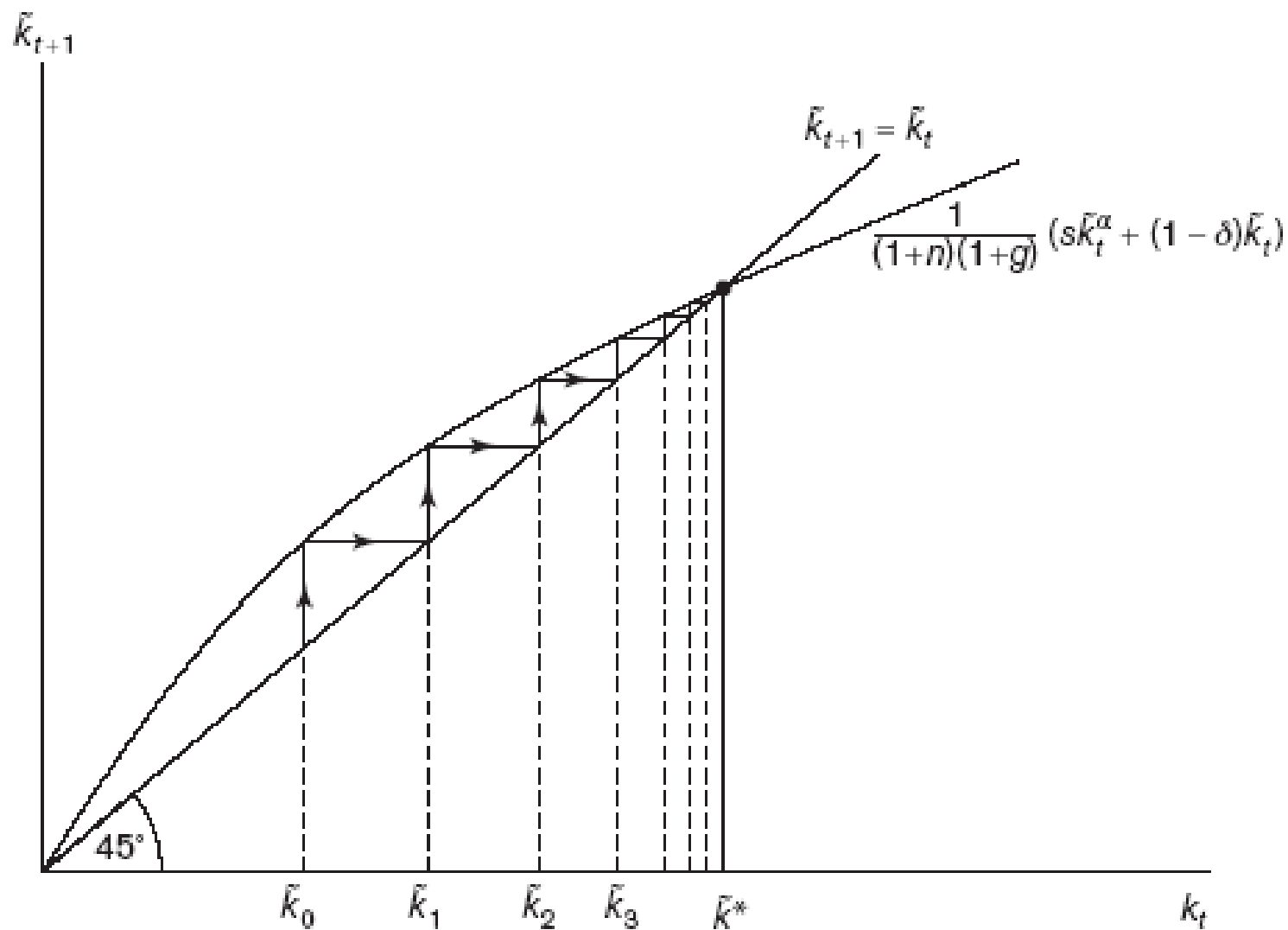
$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g)} \left(s\tilde{k}_t^\alpha + (1-\delta)\tilde{k}_t \right).$$

- It is everywhere increasing and passes through (0,0).
- The slope of the transition curve at any $\tilde{k}_t$ is:

$$\frac{d\tilde{k}_{t+1}}{d\tilde{k}_t} = \frac{s\alpha\tilde{k}_t^{\alpha-1} + (1-\delta)}{(1+n)(1+g)}.$$

- We observe: $\lim_{\tilde{k}_t \rightarrow 0} d\tilde{k}_{t+1} / d\tilde{k}_t = \infty$. Furthermore, $\lim_{\tilde{k}_t \rightarrow \infty} d\tilde{k}_{t+1} / d\tilde{k}_t < 1 \Leftrightarrow n + g + \delta + ng > 0$. We assume that the latter very plausible **stability condition** is fulfilled.

- The transition equation must then look as follows:



- Convergence of $\tilde{k}_t$ to the intersection point $\tilde{k}^*$ follows from the diagram. Correspondingly, $\tilde{y}_t \rightarrow \tilde{y}^* = (\tilde{k}^*)^\alpha$.

Some first conclusions are:

- In the long run, $\tilde{k}_t \equiv k_t / A_t$ and $\tilde{y}_t = y_t / A_t$ converge to constant levels, $\tilde{k}^*$ and $\tilde{y}^*$, respectively. These levels define steady state.
- In steady state, k_t and y_t both grow at the same rate as A_t , that is, at the rate g , and the capital output ratio, $K_t / Y_t = k_t / y_t$, must be constant.

STEADY STATE

- The Solow equation

$$\tilde{k}_{t+1} - \tilde{k}_t = \frac{1}{(1+n)(1+g)} \left(s\tilde{k}_t^\alpha - (n+g+\delta+ng)\tilde{k}_t \right)$$

together with $\tilde{k}_{t+1} = \tilde{k}_t = \tilde{k}^*$ gives:

$$\tilde{k}^* = \left(\frac{s}{n+g+\delta+ng} \right)^{\frac{1}{1-\alpha}} \Rightarrow \tilde{y}^* = \left(\frac{s}{n+g+\delta+ng} \right)^{\frac{\alpha}{1-\alpha}} .$$

- Using $\tilde{k}_t \equiv k_t / A_t$ and $\tilde{y}_t \equiv y_t / A_t$ we get the steady state growth paths:

$$k_t^* = A_t \left(\frac{s}{n+g+\delta+ng} \right)^{\frac{1}{1-\alpha}} , \quad y_t^* = A_t \left(\frac{s}{n+g+\delta+ng} \right)^{\frac{\alpha}{1-\alpha}} .$$

- Since $c_t = (1-s)y_t$,

$$c_t^* = A_t (1-s) \left(\frac{s}{n+g+\delta+ng} \right)^{\frac{\alpha}{1-\alpha}} .$$

- It also easily follows from

$$r_t = \alpha \left(\tilde{k}_t \right)^{\alpha-1} \quad \text{and} \quad w_t = (1-\alpha) A_t \left(\tilde{k}_t \right)^{\alpha}$$

that

$$r^* = \alpha \left(\frac{s}{n+g+\delta+ng} \right)^{-1} \quad \text{and} \quad w_t^* = A_t (1-\alpha) \left(\frac{s}{n+g+\delta+ng} \right)^{\frac{\alpha}{1-\alpha}} .$$

- **There is balanced growth in steady state:** k_t, y_t and w_t grow at the same constant rate, g , and r_t is constant.
- **There is positive growth in GDP per capita in steady state** (provided that $g > 0$).

STRUCTURAL POLICIES FOR STEADY STATE

- Output per capita and consumption per capita in steady state are:

$$y_t^* = A_0 (1+g)^t \left(\frac{s}{n+g+\delta+ng} \right)^{\frac{\alpha}{1-\alpha}} \quad \text{and}$$

$$c_t^* = A_0 (1+g)^t (1-s) \left(\frac{s}{n+g+\delta+ng} \right)^{\frac{\alpha}{1-\alpha}}.$$

- Golden rule: the s , that maximizes the entire path, c_t^* .
Again: $s^{**} = \alpha$.
- The elasticities of y_t^* wrt. s and $n+g+\delta$ are again $\alpha / (1-\alpha)$ and $-\alpha / (1-\alpha)$, respectively.
- Policy implications from steady state are as in the basic Solow model: encourage savings and control population growth.
- But** we have a new parameter, g (A_0 corresponds to B). We want a large g , but it is not easy to derive policy conclusions wrt. technology enhancement from our model (g is exogenous).

TAKING STOCK

- Using the transition equation and the transition diagram we showed convergence to steady state: $\tilde{k}^*, \tilde{y}^*$.
- We derived some relevant steady state growth paths, e.g.,

$$y_t^* = \overbrace{A_0 (1+g)^t}^{=A_t} \left(\frac{s}{n+g+\delta+ng} \right)^{\frac{\alpha}{1-\alpha}}$$

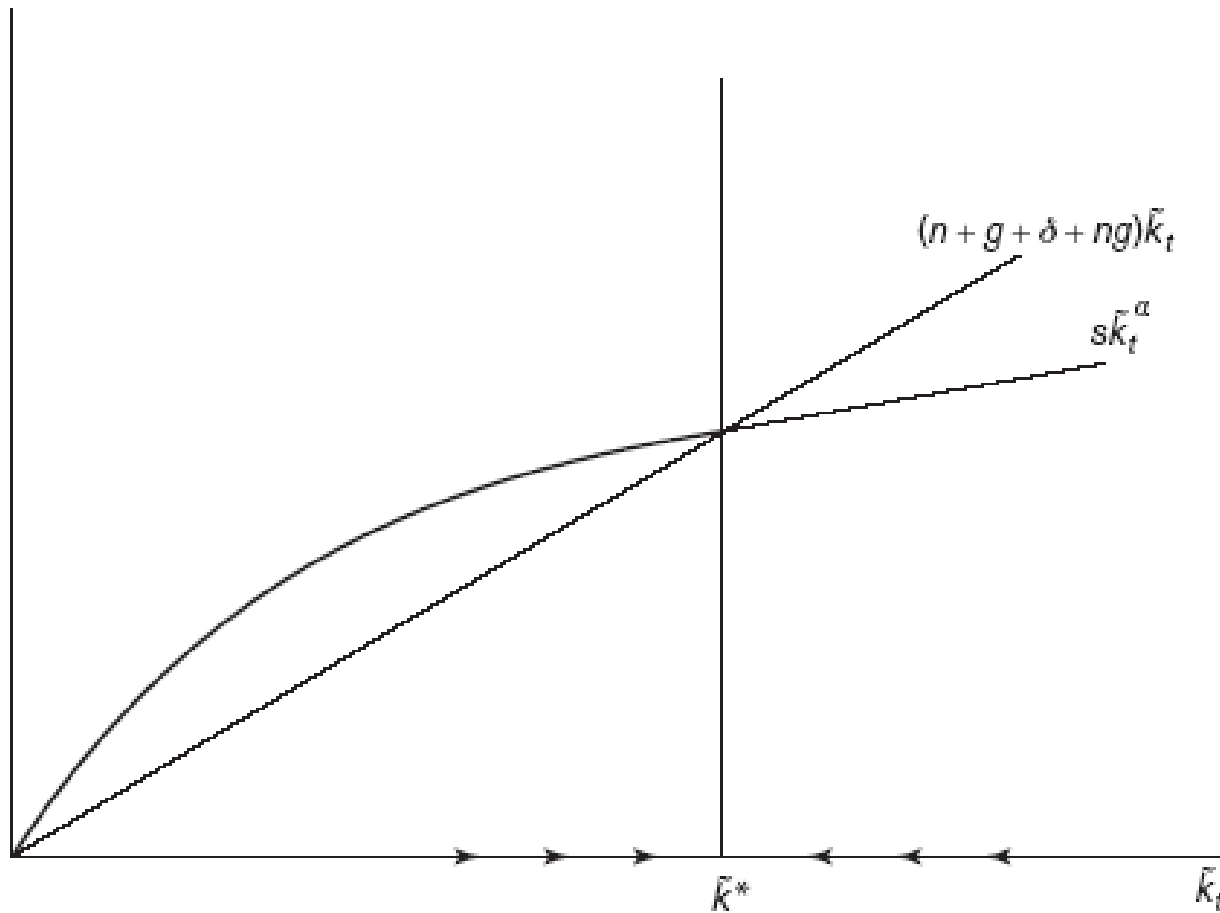
- We showed that there is balanced growth in steady state with k_t , y_t and w_t growing at the same positive growth rate, g , and with a constant real interest rate, $r^* - \delta$.
- We discussed structural policy aimed at affecting steady state.
- We showed and discussed empirics for steady state:
The model substantially underestimates the impact of the structural parameters on GDP per worker!

ADDITIONAL FEATURES

- Comparative analysis in the Solow diagrams
- The convergence process
- Growth accounting

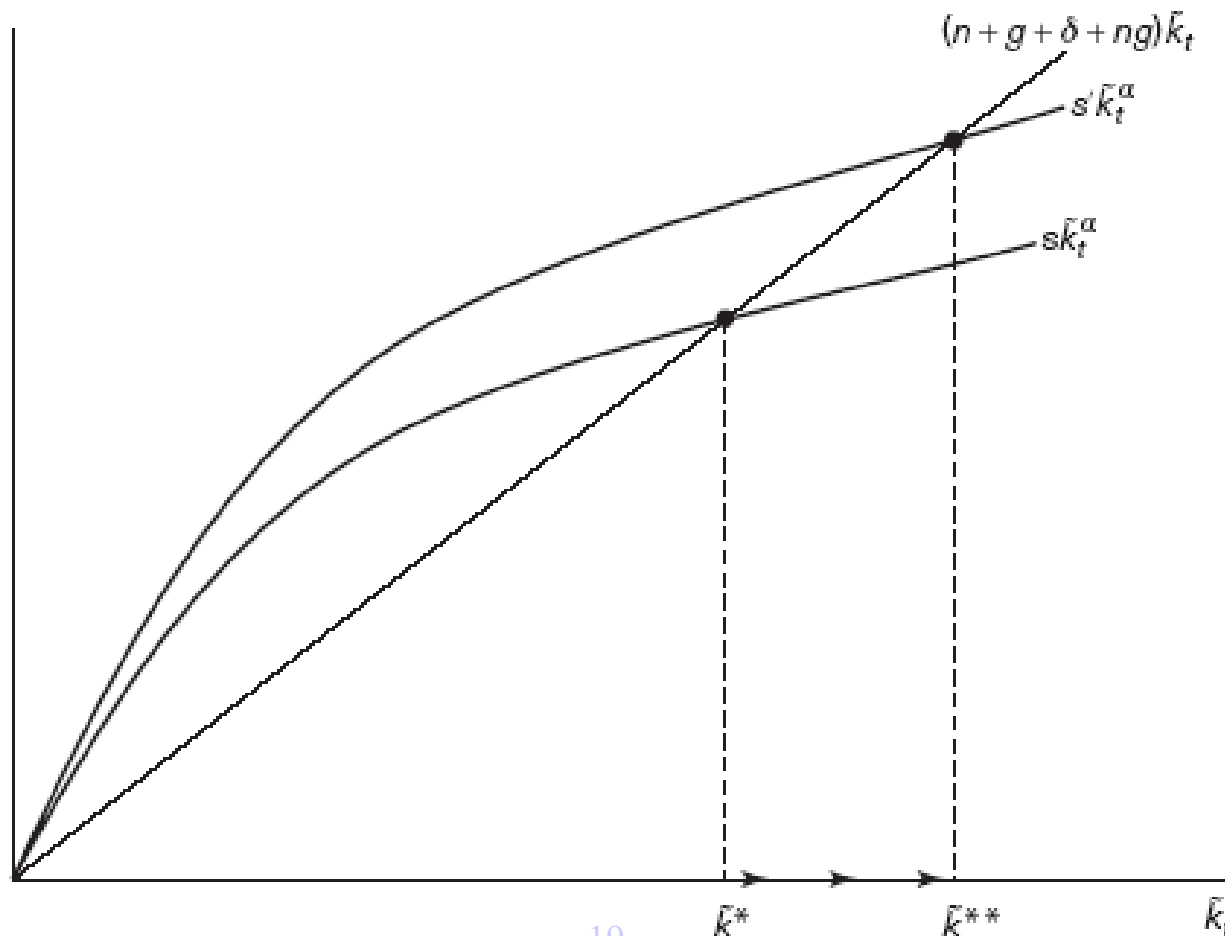
The Solow Diagram

$$\tilde{k}_{t+1} - \tilde{k}_t = \frac{1}{(1+n)(1+g)} \left(s\tilde{k}_t^\alpha - (n+g+\delta+ng)\tilde{k}_t \right).$$

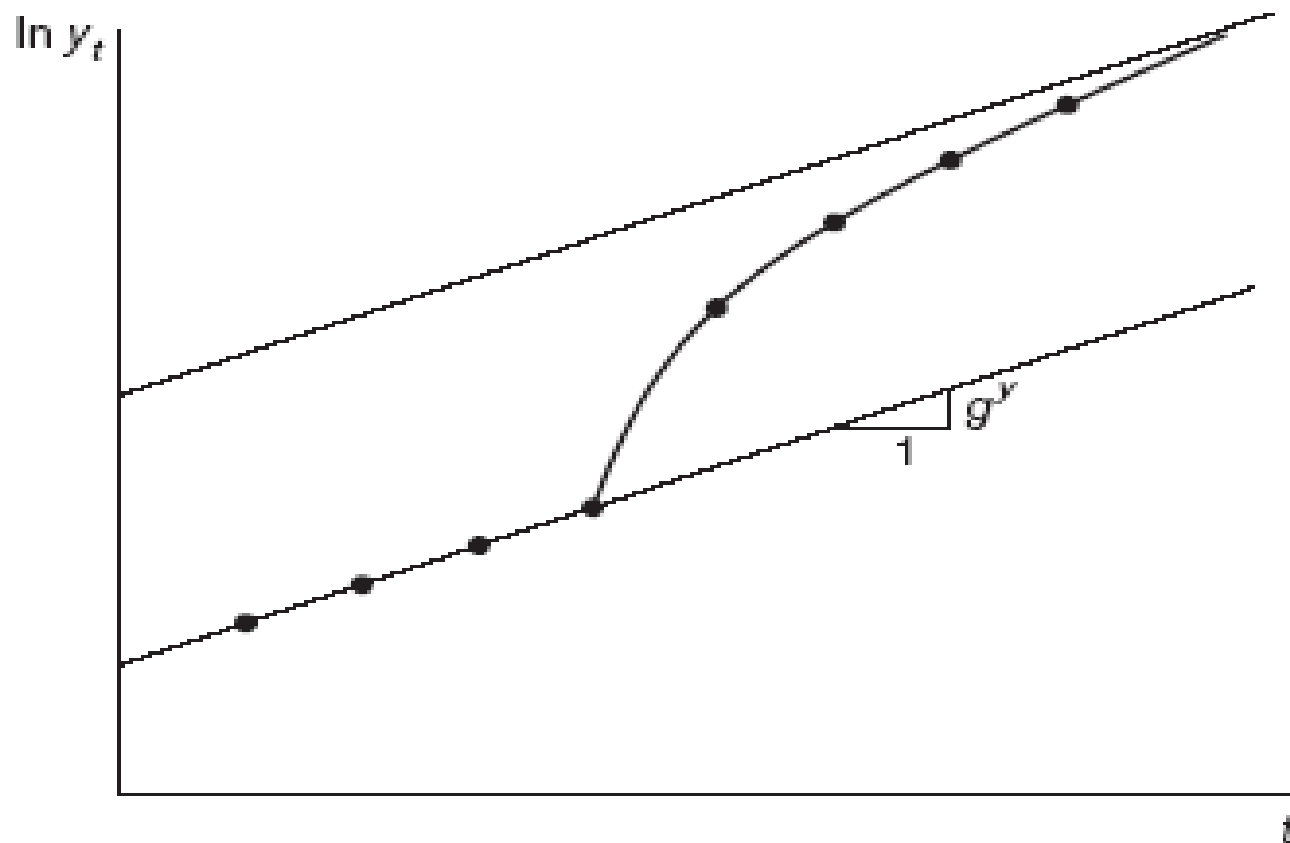


COMPARATIVE ANALYSIS IN THE SOLOW DIAGRAMS

- Initially the economy is in steady state at parameters α, s, δ, n and g . The savings rate increases permanently from s to $s' > s$.

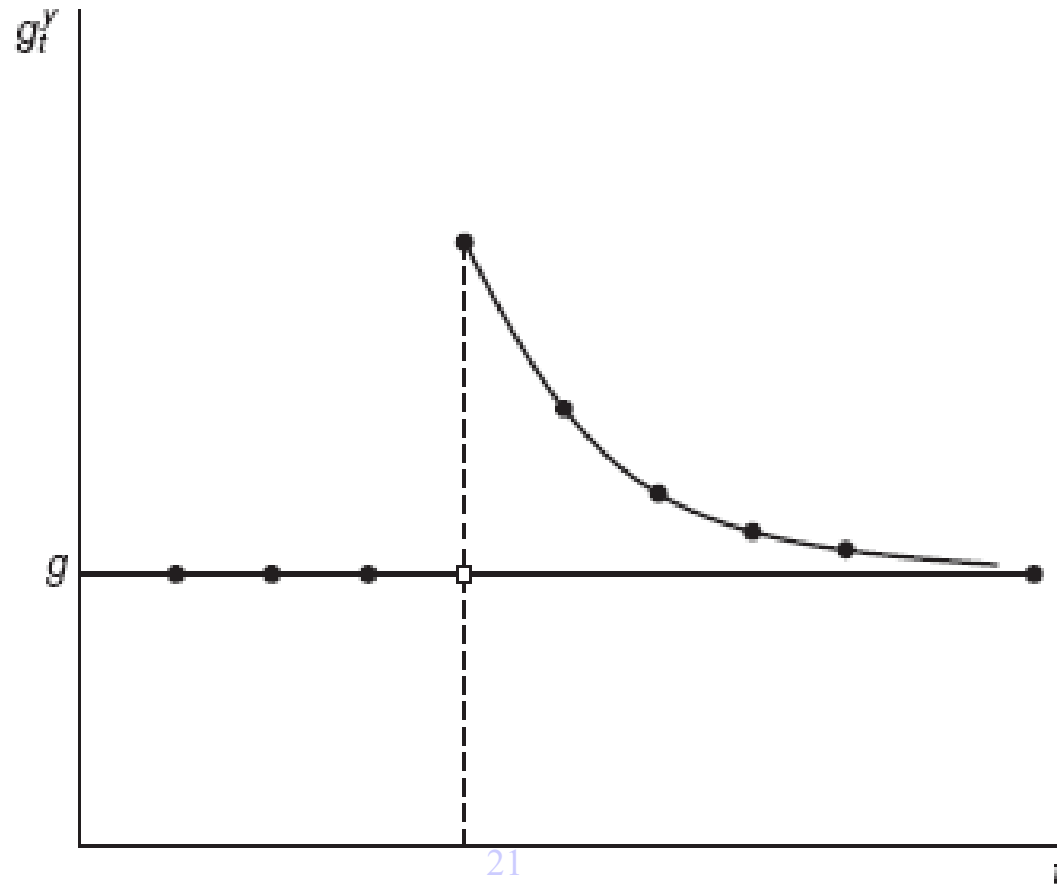


- **Old steady state:** $\tilde{k}_t = k_t / A_t = \tilde{k}^*$ and $\tilde{y}_t = y_t / A_t = \tilde{y}^*$, and k_t and y_t both grow at rate g , the lower line below:



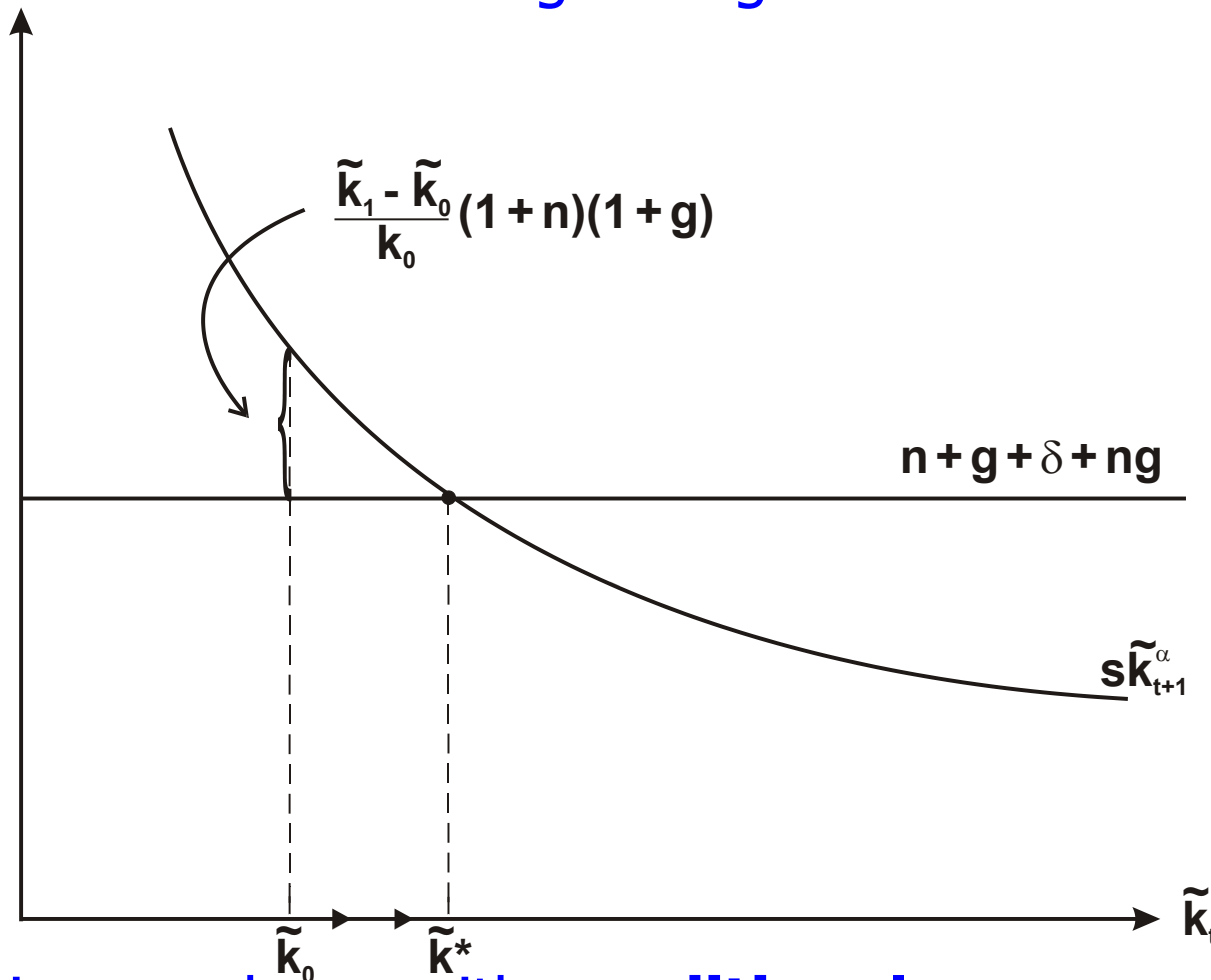
- **New steady state:** $\tilde{k}_t = k^{*'} > \tilde{k}^*$ and $\tilde{y}_t = \tilde{y}^{*'} > \tilde{y}^*$, and k_t and y_t grow at rate g but along a higher growth path, the upper line above.

- Transition:** $\tilde{k}_t = k_t / A_t$ grows from $\tilde{k}^*$ up to $\tilde{k}^{*'}$. The growth rate of $\tilde{k}_t$ jumps up and then gradually falls back to zero. From $k_t = A_t \tilde{k}_t$ follows that $g_t^k = g_t^{\tilde{k}} + g_t^A$. Hence, during the transition, k_t grows at a larger rate than g , and g_t^k jumps up and then falls gradually back to g . The growth rate of y_t jumps too, since $g_t^y = \alpha g_t^k + (1 - \alpha) g_t^A$.



CONVERGENCE IN THE SOLOW MODEL

The modified Solow diagram again:



There is accordance with **conditional convergence**:
Two countries with the same $\alpha, s, \delta, n, g, A_0 \dots$

THE PROCESS OF CONVERGENCE EMPIRICALLY

- What does the model tell us more precisely about the process of convergence , i.e., how does growth in each country depend on structural parameters and initial position?
- Using the transition equation,

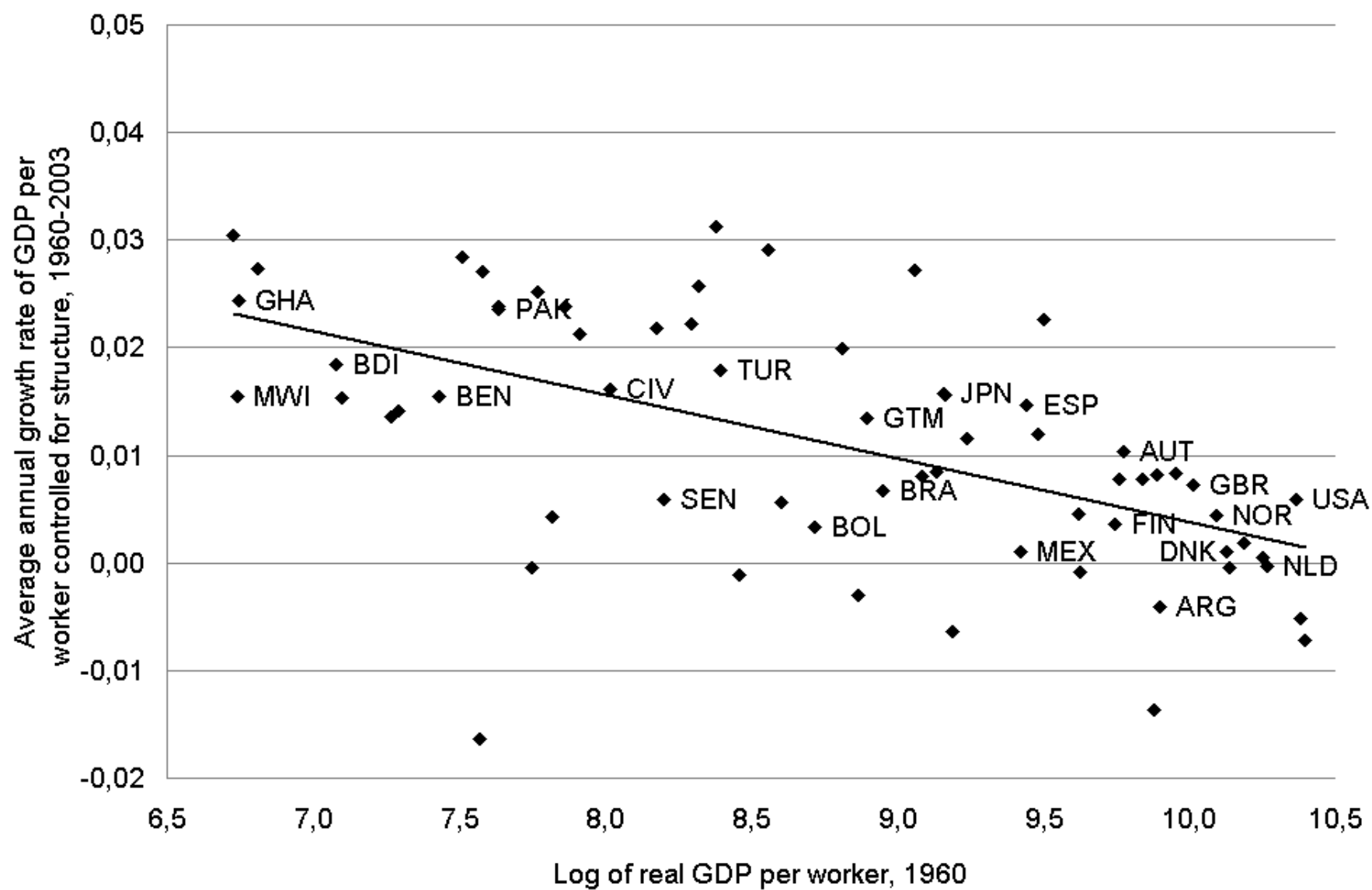
$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g)} \left(s\tilde{k}_t^\alpha + (1-\delta)\tilde{k}_t \right) \equiv G(\tilde{k}_t),$$

we derive a formula for the growth rate of y_t . To do this we **linearize around steady state**, etc.

- Assuming that g and A_0 are the same in all countries, this suggests a regression equation that gives the average annual growth rate as a function of initial GDP percapita, and transitional growth parameters (saving ratio and population growth rates).
- OLS estimation across 65 countries over the period 1960-2003 gives:

$$g_{03,60}^i = 0.063 - \underset{(0.0015)}{0.006} \ln y_{60}^i + \underset{(0.0025)}{0.016} \left[\ln s^i - \ln (n^i + 0.075) \right], \text{ adj. } R^2 = 0.38.$$

- Plotting $g_{03,60}^i - \underset{(0.0025)}{0.016} \left[\ln s^i - \ln (n^i + 0.075) \right]$ against $\ln y_{60}^i$ gives



- Positive aspects: significant parameters, relatively high R^2 , etc. This does not crucially conflict with the assumption of the same g and A_0 in all countries. But:
The model substantially overestimates the rate of convergence!
- Confronted with empirics, the Solow model does quite well, both with respect to its steady state and its convergence prediction. However, in both cases there is an empirical problem of magnitudes. It could have been better, but it's still a very well-performing model.

GROWTH ACCOUNTING

$$Y_t = B_t K_t^\alpha L_t^{1-\alpha} \Rightarrow$$

$$\ln Y_T = \ln B_T + \alpha \ln K_T + (1 - \alpha) \ln L_T \quad \text{and}$$

$$\ln Y_t = \ln B_t + \alpha \ln K_t + (1 - \alpha) \ln L_t \Rightarrow$$

$$\frac{\ln Y_T - \ln Y_t}{T - t} = \frac{\ln B_T - \ln B_t}{T - t} + \alpha \frac{\ln K_T - \ln K_t}{T - t} + (1 - \alpha) \frac{\ln L_T - \ln L_t}{T - t} \Leftrightarrow$$

- $$g_{T,t}^Y = g_{T,t}^B + \alpha g_{T,t}^K + (1 - \alpha) g_{T,t}^L$$

With data for Y_τ, K_τ and $L_\tau, \tau = t, T$ (which often is available) and with $\alpha = 1/3$, we can compute $g_{T,t}^B$ as a residual. We call this the **Solow residual**.

$$g_{T,t}^B = g_{T,t}^Y - \alpha g_{T,t}^K - (1 - \alpha) g_{T,t}^L$$

Why not growth accounting in levels?

GROWTH ACCOUNTING PER CAPITA (WORKER)

$$Y_t = B_t K_t^\alpha L_t^{1-\alpha} \Rightarrow y_t = B_t k_t^\alpha$$

$$\ln y_T = \ln B_T + \alpha \ln k_T \quad \text{and} \quad \ln y_t = \ln B_t + \alpha \ln k_t \Rightarrow$$

$$\frac{\ln y_T - \ln y_t}{T - t} = \frac{\ln B_T - \ln B_t}{T - t} + \alpha \frac{\ln k_T - \ln k_t}{T - t} \Leftrightarrow$$

$$g_{T,t}^y = g_{T,t}^B + \alpha g_{T,t}^k$$

- With data for y_t and k_t and with $\alpha = 1/3$ we can once more compute the Solow residual, $g_{T,t}^B$.
- We can use this residual to check the underlying "technological growth".

CONCLUSIONS, THE GENERAL SOLOW MODEL

- Implications for economic policies are more or less the same as those derived from the basic Solow model.
- The model implies convergence to a steady state with balanced growth and with a constant, positive growth rate of GDP per worker. Thus, the steady state prediction of the model is in accordance with a fundamental "stylized fact". However, the underlying source of growth, technological progress, is not explained.
- The steady state prediction performs quite well empirically, but the model underestimates the effect of the savings rate and the growth rate of the labour force on income per worker.
- The convergence prediction also performs well empirically, but the model overestimates the rate of convergence.